

JAPAN - KEIJINA  
INFRARED LEAGUE

Tools: Bringing Ideas to Life

Designs

- KiCad Used for develop PCB
- KiCad Used for design our components

Programmings

- VSCode Used for programming
- MaixPy IDE Used for debugg our cameras

Managements

- GitHub Used for manage our programs
- Discord Used for sharing idea or communicate

TEAM ABSTRACT

We are Team AIR. We are a team of three Ritsumeikan university students. We work daily to build robots combining precision control with high mobility for unprecedented, overwhelming performance. After securing 3rd place nationally and globally at RCJ 2025, we won 1st place in Japan at RCJ 2026. This year, we continue competing in the same league to achieve our ultimate goal: winning the World Championship. Our RCJ journey began six years ago. While we previously dedicated 10 hours a week to development during middle and high school, relocating to separate university campuses has reduced our joint activity time to about 4 hours a week. Despite this challenge, we have never given up and persistently pursue our development.

Previously, our camera-less algorithm failed at the Japan tournament due to hardware limits. Learning from this, we retained our innovative software concepts while thoroughly elevating our hardware completeness for the world stage. This hard-won experience and mindset remain deeply embedded in our team today.

Based on these experiences, our major advancements this year feature a vision-based open-source avoidance algorithm for difficult shooting angles, software that quickly adapted to rule changes and the new ball using conventional sensors, and a reliability-enhancing UI. Furthermore, we implemented a custom dribbler, a system resilient to voltage drops, and a stable, heat-free modular power supply. This poster highlights these technical breakthroughs.

OUR CONCEPTS

Previously, AIR's robots were designed around the concepts of "basics" and "stability." However, we realized that staying within this mindset kept us somewhat immature, and we felt we could never achieve overwhelming victories by simply continuing with this old concept.

BASICS

STABLE

+

DELICACY

CHALLENGE

Therefore, for this fiscal year, we decided to add the "Right Phase" to our concept, in addition to the previous "Left Phase."

At the 2025, we found that many international teams successfully used components and algorithms overlooked in Japan due to preconceptions. This inspired our "Challenge" to import and explore these new technologies. We also pursued "Delicacy"—retaining explosive power while shifting from unrefined, hyperactive movements to a controlled, powerful presence, alongside refining our base system.

To achieve this, we minimized hardware development time by avoiding drastic changes, shifting our schedule to allocate maximum time to programming. Therefore, this year's robot is based on our Japan Championship machine, with the vast majority of parts standardized between both units.

ATK & DATA

DF & DATA

ROUGH SKETCH

Main Unit

- Line Photo IC B19H1LS 3x 50
- IR Ball Sensor TSSP4038 3x 24
- Ultra Sonic Sensor RCW-0001 3x 4
- CPU - RP2350B
- CPU - RP2350A
- CPU - XIAO ESP32-S3
- CPU - XIAO ESP32-S3
- UnitV cam 3x 3

Battery Unit

- DCDC - AP445008P
- LiPo - 3S 2200mAh
- ESC - 20A
- MD - DRV8432 x 2
- BOOST - XL6009
- BLDC - YSIDO 1410 2580KV
- Motor - Maxon DCX x 4 / Maxon RE16 x 4
- Solenoid - CB1037

Our Robot's COST (sponsored item includes) main parts: \$2576 small parts: \$2576 TOTAL COST: \$2576

PARTS & PCB DIAGRAM

MAIN & UI BOARD

Parts : Teensy4.1 / BNO055 / ESP32S3 / UIs

We selected the Teensy 4.1 as our main microcontroller because it offers many UART ports and simple debugging. Additionally, we equipped two ESP32S3 sub microcontrollers for communication and UI processing. These components have increased the robot's real-time processing capabilities, leading to a much faster performance.

BALL SENSOR BOARD

Parts : RP2350A / TSSP58038

We use the RP2350A as our sub microcontroller. Although the community claims the new IR ball cannot be detected by conventional sensors, we overcame this challenge. To achieve this, we upgraded from the ATmega32U4. It communicates with the main microcontroller via UART.

LINE SENSOR BOARD

Parts : RP2350B / B19H1LS / LM393

We use the RP2350B as the line sensor sub-microcontroller for its abundant I/O, outputting digital signals. Leveraging our 2025 World Championship experience, we incorporated a PWM illumination adjustment system to ensure recognition in any environment. Communication with the main microcontroller is via UART.

POWER DISTRIBUTION BOARD

Parts : AP64500SP / 10A Fuse / Battery LCD / Pch MOSFET

It steps down LiPo battery voltage to 7V and 3.3V using DC-DC converters, ensuring stable power to each PCB. A 10A fuse protects the circuit from overcurrent. Unlike previous designs with severe heat generation and power loss, we meticulously optimized both high-efficiency IC selection and PCB layout design to minimize thermal dissipation and power loss.

MOTOR DRIVER BOARD

Parts : DRV8432 / YSIDO ESC20A

We chose the DRV8432 motor driver for its LAP support, 500kHz frequency, and high current capacity. This modular design ensures easy replacement. Additionally, the highly extensible interface seamlessly accommodates either a standard motor driver or an ESC for the dribbler.

BOOST & KICKER BOARD

Parts : XL6009 / Nch MOSFET / Pch MOSFET / AQW212

The XL6009 module steps up 12V to 46V, utilizing N/P-ch MOSFETs for solenoid switching. Previously, a P-ch MOSFET limited the solenoid's power. To overcome this, we adopted a high-performance N-channel MOSFET to maximize kick power. To protect the microcontroller from this high-voltage signal, the AQW212 photocoupler provides electrical isolation.

OVERCOME 2026 RULE CHANGES: A PROVEN IR SENSOR SOLUTION

For the 2026 World Cup, the IR ball regulations shifted from Mode A to Mode D pulses, and the physical size was halved. While the Japanese robotics community considered conventional sensor designs unusable under these new constraints, we challenged this assumption. We successfully achieved highly accurate distance and angle tracking using the same sensor and circuit configuration from our Japan Open robot. True to the RoboCup spirit of open source, we share our complete design—composed of the following four elements—to support the community's future:

— ANGLE CALCULATION via DIGITAL WEIGHTING

- 7-Sensor Spatial Window: Selects the sensor with the highest pulse count along with 3 adjacent sensors on both sides (total 7 sensors) to bound the target area.
- Square-Weighted Center of Mass: Drops any sensor data below 25% of the maximum peak value to eliminate stray light. Then, vector synthesis is performed using the square of the pulse counts as weight coefficients ( $w_i = \text{count}_i^2$ ) to sharpen directional resolution.
- Vector-Domain EMA: Applies a 10% Exponential Moving Average ( $\alpha = 0.10$ ) to the synthesized X and Y coordinates prior to the atan2f calculation, preventing angle chattering caused by transient sampling noise.

$$w_i = \begin{cases} (C_i)^2 & (C_i \geq 0.25 \cdot C_{max}) \\ 0 & (\text{otherwise}) \end{cases} \quad X = \sum_{i \in W} w_i \cos \theta_i, Y = \sum_{i \in W} w_i \sin \theta_i$$

$w_i$  : Weight coefficient for sensor i based on pulse intensity.  $C_i$  : Raw pulse count of sensor i\$ obtained from 500x sampling.  
 $C_{max}$  : Maximum pulse count among all 24 sensors.  $W$  : Spatial window consisting of 7 sensors centered around the peak sensor.  
 $\theta_i$  : Physical placement angle of sensor i (intervals of 15 degrees). X,Y: Coordinates of the synthesized directional vector.

— DISTANCE SENSING via PULSE WIDTH

- 500x Oversampling & 12-Frame Temporal Buffer: Performs 500 direct GPIO reads per sampling loop to detect micro-pulses. The results are stored in a 12-frame moving average buffer for time-domain noise reduction.
- Top-12 Spatial Integration: Identifies the peak sensor and averages the time-smoothed values of the 12 closest surrounding sensors. This spatial integration effectively averages out localized floor reflections and signal scattering.
- Dynamic Hysteresis Filter: Implements a dual-mode thresholding system. If the variance between the new raw distance and the previous state is under 20, the output is locked to prevent jitter. If it exceeds 20, the system transitions to tracking mode using a 20% blend rate to follow rapid ball movements.

$$S_i = \frac{1}{12} \sum_{k=0}^{11} C_i[t - k]$$
$$D_{raw} = \frac{1}{12} \sum_{d=-5}^6 S_{max+d}$$

$S_i$ : 12-frame temporal moving average of sensor i at time t.  $C_i[t-k]$ : Raw pulse count of sensor i recorded k frames ago.  
 $D_{raw}$ : Raw distance integrated from the 12 spatial sensors around the peak.  
 $S_{max}+d$ : Smoothed pulse count of the sensor located \$d\$ steps away from the peak.

A program to retrieve this ball distance has been uploaded to GitHub. You can access it via the link on the right.

— MCU UPGRADE from ATMEGA32U4 to RP2350A

To implement these 2 system, the previously used ATmega32U4 (8-bit, 16MHz) fell critically short in both processing power and I/O pin count. Therefore, we adopted the RP2350A, high-performance microcontroller. The key reasons for this choice are threefold:

| Features    | Data Capture                                   | Processing                                                           | Circuit Design                                                     |
|-------------|------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------|
| ATmega 32U4 | Sequential pin reading                         | High latency in heavy math                                           | Requires multiplexers                                              |
| RP2350A     | Simultaneous 24-pin capture via SIO in 1 cycle | 150MHz+ FPU: Runs vector synthesis & 12-frame averages easily in 1ms | Direct 24-pin connection with abundant native GPIOs (Zero latency) |

— REDESIGN: IR RECEPTION BOUNDARY CONTROL

This system requires uniform IR pulse reception. Due to the smaller ball size hindering close-range detection, we implemented two hardware optimizations:

- 15cm Enriched Board Diameter (Up from 10cm): Prevents the robot frame from blocking IR signals, eliminating blind spots for stable angle and distance tracking across the entire field.
- "Narrow & Deep" Sensor Shielding: Narrowed the slit width by 1cm and recessed the sensors 2cm backward. This sharpens directional selectivity and drastically eliminates false positives from stray reflections.

RELIEABLE CIRCUITS

Just before the 2026 Japan Open, power noise and GND instability (fig1) caused unexpected microcontroller resets. To resolve this, we implemented the following optimizations in our new circuit:

— STABLE & ROBUST GND DESIGN

- Implemented single-point grounding
- Maximized continuous ground to lower impedance
- Thoroughly eliminated all ground loops
- Removed islands to eliminate noise sources

— MORE VOLTAGE STABILIZATION

- DON'T trust DC-DC voltage
  - Having switching Noise
- Using Linear Regulator
  - Clean, Noise-free supply
- 7V DC-DC → 5V LDO → MCU

— AVOID HUMAN ERRORS

- Use Colored Connectors
- Make Colored Wires
- Silkscreen Specifications
- Onboard UIs & LEDs
- More Accessible Test Points

ADVANCED OMNI-WHEEL

At the 2025 World Championship, we could not move as desired due to a lack of wheel grip. To resolve this, we fabricated single-layer thin and thick rubber wheels, spiked wheels, and double-layer thin and spiked wheels, and conducted an experiment to compare their respective grip forces.

Since simply rubbing the wheels on the carpet by hand was insufficient to measure the exact increase in grip, we devised a method to quantitatively evaluate it. We used the BNO055 to estimate the robot's acceleration and plotted the collected data on a graph for precise, data-driven analysis.

EXPERIMENT & RESULTS

We averaged 10 startup acceleration trials per wheel. Because wheel type affected acceleration, we adopted dual-layer urethane omni wheels, whose superior grip significantly benefits our robot.

FEINT-BASED AVOIDANCE SYSTEM

Solid defense is now standard, but conventional avoidance lacks certainty. Our feint system works without a dribbler on any vision robot—likely first in IR/Vision leagues. While we added a dribbler for match stability, this proves precise control can beat hardware limits. After ball capture, the system triggers two phases based on goal distance:

PHASE 1: CLOSE PROXIMITY (ENEMY-BASED)

- 1 Detect the two largest blocks in the image in order
- 2 Drive towards the back from the center of the largest block

PHASE 2: LONG DISTANCE (FEINT-BASED)

- 1 Drives into the narrower goal area and attract the enemy
- 2 Drive to the other side as the narrow area disappears

THE LOGIC BEHIND THE STRATEGY: WHY OUR FEINT-BASED CONTROL IS HIGHLY EFFECTIVE?

— EXPLOITING ROBOTIC PHYSICAL LIMITS

Our sudden direction changes force the defender to face critical hardware bottlenecks, such as sensor processing delays and tire slippage on the carpet. This makes it physically impossible for the opponent to react in time.

— ALGORITHMIC DECEPTION

By intentionally driving toward the narrower goal space, we deliberately manipulate the opponent's tracking algorithm. This movement tricks the defender into moving outward, inevitably opening a massive defensive gap in the center.

LiPo & SPEED DROP COMPENSATION

During the 2025 season, we discovered that the robot's movements differed between testing and actual matches due to a decrease in motor rotation speed caused by a drop in battery voltage. The method to resolve this is described below.

— Option 1: Time-based control

First, we devised a time-based system increasing speed to compensate for discharge. However, it was incompatible with power-off states, and motor usage was inconsistent.

— Option 2: Voltage-based control

Secondly, a voltage-based system was devised to compensate for dropping motor voltage due to power supply depletion. However, required circuit modifications were a concern.

We adopted option 2 for stability, using a resistive divider to read the supply voltage percentage as an analog input to the Teensy 4.1.

Experiment: Measure and compare robot travel time across the court at various supply voltages.

These results confirm the high effectiveness of this control system, which successfully maintained a constant speed regardless of the voltage level.

RCJ COMMUNITY OUTREACH

After the 2025 world championship, we focused not only on robot development but also on promoting RCJ and education in Japan.

- Exhibitions & Events: Showcased the robot at public tech exhibitions to inspire the community.
- Media Features: Promoted RCJ and engineering through TV and newspaper appearances.
- STEM Education: Organized and led hands-on robotics workshops for elementary school students.

HYPER-FAST WALL CONTROL

We aimed to replicate the tactic of the RCJ 2025 World Champions, "Lovbot Legend," who decelerate near lines to prevent out-of-bounds penalties. Though they used ultrasonic sensors, this method has two drawbacks: slow processing (50ms latency) and false positives from opponent robots. To overcome these challenges, we developed the following system:

— Non-Blocking Control via FreeRTOS:

Utilized the dual-core architecture of the Seeed XIAO ESP32S3 with FreeRTOS to execute tasks at a 10–15 ms cycle, effectively bypassing the blocking limitations of the pulseIn() function and mitigating sensor latency.

— Advanced Object Discrimination

Eliminated false positives by analyzing distance transition trends to accurately differentiate between walls and opponent robots.

EXPERIMENT:

Testing its ability to overcome ultrasonic latency, we compared out-of-bounds errors on slippery, low-visibility fields across three conditions: No Ultrasonic, Pre-sensing, and Post-Optimization.

RESULTS:

As shown in the graph on the right, out-of-bounds errors dropped significantly, proving reliable line prevention. This innovative breakthrough successfully overcame our technical weakness.

ROBOT ALGORITHMS

Uses the ball's angle and distance to flank it via the shortest path, whether still or moving.

After capturing, the robot aligns itself toward the goal and kicks, thereby evading enemy robots.

The robot traces along the line to transport the ball, while remaining resilient against physical pushing.

The robot can force the ball into the goal area, directly leading to a score.

The robot tracks the line near the goal, and it also works well on diagonal paths.

When the ball stays in front for a set time, the robot kicks it to avoid enemies.

When off the line, the robot goes back to the goal, which keeps it court, the attacker changes its role via Communication.

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